

USER MANUAL

For,
TRACK BASED LUBRICATORS
(ELECTRONIC TYPE)
Model: TBL-ELEC.



BY:



CHAKRADHAR INDUSTRIES LLP

Office no.1, bhuvaneshwari building

sahaji raje road, vile parle east

mumbai 400057, india

TEL: +91 22 26248057/ 65735856

FAX: +91 22 26706817

Email: info@chakradharindustries.com

Web: www.chakradharindustries.com

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1. Scope of Application

With the development of railways and large increase in speed and frequency of operations, the wear of rail curve caused by train is a critical problem, especially for small radius curves. In order to prolong the service life of flange and rail, our company has successfully developed the Track Based Lubricator: Electronic Type, which is suitable for the application of special anti-wear (lubricant) on the rail section and edge of the wheel. The system is able to automatically apply grease on rail on specific count of wheels set on controller device of the entire control system.

2. Main Technical Specifications:

1. Applicable train speed: 10 to 200km/h
2. Coat coverage: Above 1 km.
3. Storage tank capacity: 50 kg & 20 Kg
4. Working mode: Automatic.
5. Operating Temperature: -10 to +75 °C

3. Part Specification and its Uses:

1. Solar Panel Details:	a. Peak power rating: 300W x 2 b. Peak operating current: 8A c. Peak operating voltage: 35.58V d. Short circuit current: 7.29A
2. Sensor:	a. Type: Inductive proximity. b. Operating Voltage: 10-30V c. Rated Current: 0.25A d. Detection distance: 50mm.
3. Electric Grease Pump:	a. Operating voltage: 24V DC b. Rated current: 10A
4. Programmable Counter:	a. Input Voltage: 230V AC b. Power Consumption: 2W @ 230V AC c. Working temperature: 0 to 55 °C d. Operating humidity: 95% RH (non-condensing)

4. Operating Manual

The machine is composed of control panel which is controlled by a Programmable Counter device on 230V AC supply energy supply. Programmable Counter device consist of a seven segment display which indicates cycles and other information about the machine such as serial number. The machine operated on 240V AC supply which converted to 24V DC through a Switched Mode Power Supply (SMPS). The system can sustain voltage up to 48V but it is recommended to keep the internal power supply below 30V DC.

5. Working Principle

As the specific wheels pass through the detection area, the proximity sensor gives the signals to the programmable counter where counter counts the each count and activates the output of counter which is Electric Motor. As the motor is turned on the piping system of motor delivers the grease to the distribution block which is connected to the grease applicator which applies the grease to wheel flange and the edges of rail.

The programmable controller takes digital input from proximity sensor, the counter counts the signal and activates the motor depending on preset value of counter for preset value of time in seconds. The values of counter can be set manually.

6. INSTALLATION GUIDE:

- ❖ Check for loose connection of components.
- ❖ The machine is fixed to the building clearance to ensure safety.
- ❖ Build a reinforced concrete structural platform within the limits of the building, with four fixed footings on the top of the platform, which are firmly fixed with the machine.
- ❖ Fix the coating plate to the curve, and act on the side and lock it.
- ❖ Install the sensor device to the front of the coating plate and adjust the distance from the eaves height below to rail gauge face.
- ❖ Put the Grease pump directly into the anti-friction composition tube, it can do lubricating works. If the lubricating oil is not smooth, use the stick to the direction of the pump head, insert a few times to and remove air.

7. MAINTENANCE POINTS:

- ❖ Do not extend your hand into the moving parts during the operation of the machine.
- ❖ The connecting parts of the tubing should be tightened and strengthened to avoid accidents caused by the loosening of the train.
- ❖ Once the oiled board is clogged, remove and clean up the oil stain and reinstall.
- ❖ Any found leakages through hose pipe must be replace by replacing new authorized parts only.
- ❖ If oil is not get circulated then do manual pumping through till outlet of the actuator working starts.

8. SERVICE CHECK SHEET:

- ❖ Check whether Lubricator Cabin must set firm on the platform to sustain over train vibrations and environmental condition.
- ❖ Check Sensor level as it should be fit as per provided level to sense wheel edge.
- ❖ Check Applicator ports shall be open after all the connections are fitted internally before start the machine.
- ❖ Check Applicator height should not be more that rail face.
- ❖ It must be mounted firmly with rail base with profile mounting clamps.
- ❖ Do manual check to circulate grease throughout the system.
- ❖ Check Applicator hose pipe connection, it must be tight and check all ports passages are clear.

9. OTHERS:

1. About standard accessory, refer to the packing list.
2. Our company is responsible for maintaining and replacing the product being failed to use due to quality reason.

10. PACKING LIST:

Name	Track based Lubricator (Electronic Type)	Model	TBL-ELE
No.	Item	Quantity	
1	User Manual	1 pc	
2	Pre Dispatch Certification	1 pc	
3	Allen wrench	1 pc	
4	Allen wrench	1 pc	
5	Solid wrench	1 pc	

selec

XC2200/XC3200, XC2200-C/XC3200-C

Operating Instructions



72 x 72

96 x 96

SPECIFICATIONS

Display	6 (Red) + 4 (Green) digits, 0.43" height, 7 seg LED LED's : 2(Red) +2(Green)
No. of keys	3
Supply Voltage	90 to 270 VAC, 50/60Hz
Sensor Supply	12VDC, 30mA (±10%) (Short circuit protected)
Range	Total : Least count 0.0001, 0.001, 0.01, 0.1, 1 Rate : Auto ranging. RPS, RPM or RPH mode user selectable (minimum frequency for rate mode is 4 RPM)
Set Points	Two set points, each programmable from 0.0001 to 999999
Rate (Rt) / Batch(Bt) Selection	User selectable Rate / Batch at lower display
Relay 2 operation modes	User selectable Relay 2 operation 1) Operate on Set2 2) Operate on Set Batch (SPBT)
Operating modes	On delay, Interval, Time pulse repeat, Auto reset
Counting modes	Unidirectional, Quadrature, Bi - directional Directional (Up / Down)
Accuracy	Rate : 0.05% ± 2 counts
Count Inputs	5 to 30VDC from proximity switches, encoders, solid state devices, potential free contacts like limit switches, relays
Maximum Input Frequency	30Hz, 5000Hz
Scale Factor	Programmable form 0.00001 to 9.99999 x 10 ⁿ Where n = -5, -4, -3, -2, -1, 0, 1, 2
Reset Input	1) Front panel (user selectable) 2) Remote reset Minimum reset time for remote reset is 10ms
Relay Output	2 relays, 1C/O each rated 5 A @ 230 VAC / 24 VDC
Memory Retention	10 years
Configuration Lock	Via rear terminals to avoid inadvertent change in configured parameters
Housing	Flame retardant plastic

ORDER CODE INFORMATION

Product	RS485 Communication	Certification	
		CE	UL LISTED
XC2200	—	—	—
XC2200-C	■	—	—
XC3200	—	—	—
XC3200-C	■	—	—
XC2200-CE	—	■	—
XC2200-C-CE	■	■	—
XC3200-CE	—	■	—
XC3200-C-CE	■	■	—

ENVIRONMENTAL CONDITIONS

Temperature	Operating : 0 - 50°C, Storage : -20 - 75°C.
Humidity	95% RH
Mounting	Panel Mounted
Weight	XC2200/XC2200-C : 228gms XC3200/XC3200-C : 263gms

COMMUNICATION (Only applicable to XC2200-C & XC3200-C)

Communication port	RS 485 Slave
Communication Protocol	MODBUS RTU
Baud Rate	9600, 19200, 38400, 57600, 115200 (User configurable via software and hardware) Default= 19200
Default setting	Baud Rate : 19200 Parity : None Stop bit : 2

SAFETY PRECAUTIONS

This manual is meant for personnel involved in wiring, installation, operation and routine maintenance of the equipment. All safety related conditions, symbols and instructions that appear in this operating manual or on the equipment must be strictly followed to ensure operator and instrument safety. Any misuse may impair the protection provided by the equipment.

CAUTION : Read complete instructions prior to installation and operation of the unit.

CAUTION : Risk of electric shock.

INSTALLATION INSTRUCTIONS

1. This equipment, being built-in-type, normally becomes a part of the main control panel and the terminals do not remain accessible to the user after installation.

2. Conductors must not come in contact with the internal circuitry of the equipment else it may lead to a safety hazard that may endanger life or cause electrical shock to the operator.
3. Circuit breaker or mains switch must be installed between the power source and supply terminals to facilitate power 'ON' or 'OFF' function.
4. The equipment shall not be installed in environmental conditions other than those specified in this manual.
5. Since this equipment forms part of the main control panel, its output terminals get connected to the host equipment. Such equipment shall also comply to EMI / EMC and safety requirements like CE standard procedure.
7. Thermal dissipation of equipment is met through ventilation holes provided on housing of equipment. Obstruction of these ventilation holes may lead to a safety hazard.
8. The output terminals shall be loaded strictly as per the values / range specified by the manufacturer.

ELECTRICAL PRECAUTIONS DURING USE

Electrical noise generated by switching of inductive loads can create momentary disruption, erratic display, latch up, data loss or permanent damage to the instrument.

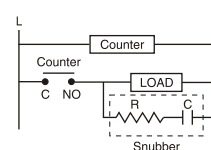
To reduce noise :

Use of Selec make Snubber across load is recommended.

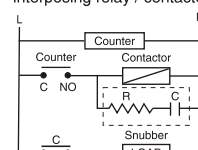
NOTE : Below mentioned diagram is applicable only for 230V relay outputs.

TYPICAL CONNECTIONS FOR LOADS

For load current < than 0.5A



For higher loads use interposing relay / contactor

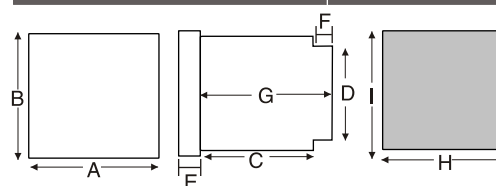


NOTE : a) Use snubber as shown above to increase life of internal relay
b) Use separate shielded wires for inputs.

MECHANICAL INSTALLATION

Outline Dimensions (in mm)

Panel Cutout (in mm)



MODELS	DIM	A	B	C	D	E	F	G	H	I
XC2200/XC2200-C		72	72	74	61	5.5	3	91	67	67
XC3200/XC3200-C		96	96	45.5	90.5	3	13	55	90.5	90.5

For installing the controller

1. Prepare the panel cutout with proper dimensions as shown above.
2. Fix the unit into the cutout. Insert the clamp from both sides and tighten the screws.

CAUTION

The equipment in its installed state must not come in close proximity to any heating sources, caustic vapors, Oils, steam, or other unwanted process by products.

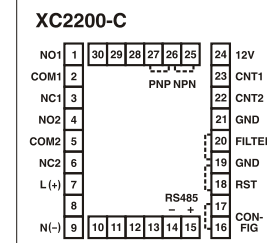
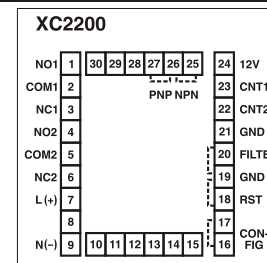
EMC Guidelines :

1. Use proper input power cables with shortest connections and twisted type.
2. Layout of connecting cables shall be away from any internal EMI source.

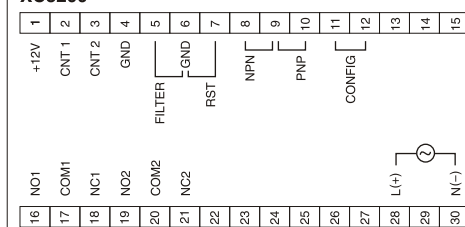
MAINTENANCE :

1. To avoid blockage of ventilation holes, clean the equipment regularly using a soft cloth.
2. Do not use Isopropyl alcohol or any other organic Solvents for cleaning.

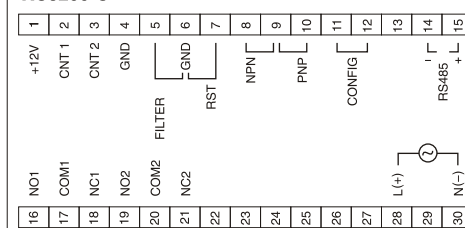
TERMINAL CONNECTIONS



XC3200



XC3200-C

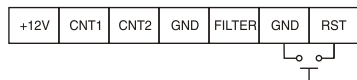


Product		Terminals
XC2200/ XC2200-C	XC3200/ XC3200-C	
24	1	+12VDC (sensor supply)
23	2	CNT1
22	3	CNT2
21	4	GND
19 - 20	5 - 6	RST
18 - 19	6 - 7	FILTER
25 - 26	8 - 9	NPN
26 - 27	9 - 10	PNP
16 - 17	11 - 12	CONFIG
14 - 15*	14 - 15*	RS485 (- & +) *Only applicable to XC2200-C & XC3200-C
1 - 2 - 3	16 - 17 - 18	NO1 - COM1 - NC1
4 - 5 - 6	19 - 20 - 21	NO2 - COM2 - NC2
7 - 9	28 - 30	L (+) & N (-)

RESETTING

By front key : Press RST key continuously for 3 sec. The counter resets and starts counting again, and the display shows the rate or total as per selection.

Remote reset : The unit can be reset from a remote push button as per connections shown in the figure :



CONFIGURATION SCHEME (Parameter setting)

- 1) Short terminal [CONFIG] no 16, 17 for XC2200 / XC2200-C & 11, 12 for XC3200 / XC3200-C
- 2) Turn power on
- 3) Program settings as per instructions given below :
- 4) Value of parameter is stored after pressing Rst key.

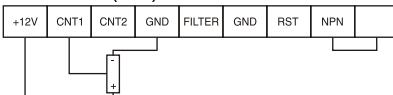
Config. Parameter	Display	Description
Scale factor mantissa	Factory setting : 1.00000 *Blinking digit 	Key : Increments the blinking digit after every key press and rolls over from 0 to 9. Key : Shifts the blinking to next digit (right) for every key press. Range : 0.00001 to 9.99999
Scale factor exponent	Factory setting : 0 *Blinking digit 	Press Exponent = 0 i.e 10 ⁰ Range : 0 to 2, -5 to -1
Resolution	Factory setting : 1 Blinking 	Press Least count = 1 Range : 0.0001 to 1
Counting mode	Factory setting : Unidirectional Blinking 	Press Uni direction Range : Unidirection, Bidirection, Quadrature NOTE : Quadrature mode maximum speed - 2kHz

WIRING DIAGRAM FOR COUNT INPUT

(Sensor color codes :-

Red = +12V, Green = CNT, Black = GND
Brown = +12V, Black = CNT, Blue = GND)

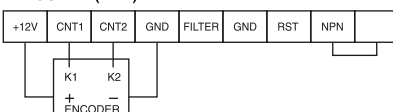
PROXIMITY (NPN)



CONTACT / SWITCH



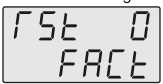
ENCODER (NPN)



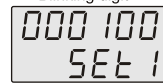
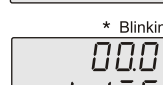
NOTE :

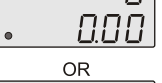
1. For PNP sensors, connect 1kohm resistor between CNT and GND terminals
2. If the sensors (Proximity / Encoder) require more than 30mA current, use external power supply to power the sensors.

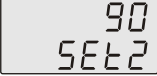
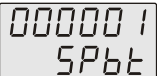
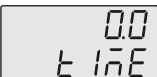
Config. Parameter	Display	Description
Count direction	Factory setting : Up Blinking 	Press Up count Range : UP count, Down count
Relay Operating mode	Factory setting : On delay Blinking 	Press On delay Range : On delay, Interval
Reset mode	Factory setting : None Blinking 	Press None (no Autoreset, no time pulse repeat) Range : None, Autoreset, Time pulse repeat.
Input frequency	Factory setting : 5000Hz Blinking 	Press 5000Hz Range : 5000 Hz, 30 Hz
Rate unit	Factory setting : RPS Blinking 	Press RPS, Rate per second Range : Rate per second, Rate per minute, Rate per hour
Overrun / non overrun	Factory setting : Over run Blinking 	Press Over run Range : Overrun, Non over run
Rate / Batch Selection	Factory setting : Rate Blinking 	Press Lower display Rate value Range : Rate, Batch
Relay 2 Operation Selection	Factory setting : SET2 Blinking 	Press Operation based on Set2 Range : Operation based on SET2, Operation based on Set Batch
Communication setting	Factory setting : BR19K2 & 1 Blinking 	Press Baud Rate Range : 9600, 19200, 38400, 57600, 115200 Key : Increments the blinking digit after every key. press and rolls over from 0 to 9 Key : Shift the blinking to next digit (right) for every key Press Slave Id from 0 to 255 Note : Only applicable to XC2200-C & XC3200-C
Front panel reset	Factory setting : enabled Blinking 	Press Reset key enabled Range : Yes, No

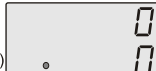
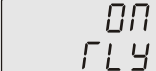
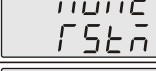
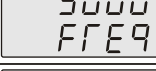
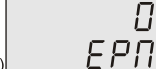
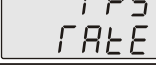
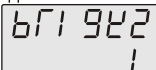
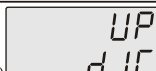
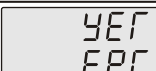
Config. Parameter	Display	Description
Reset cumulative total	Factory setting : Rtot 0 Blinking 	Press   No action from rtot 0 - 4 Range : R tot 0, R tot 1, R tot 2, R tot 3, R tot 4, R tot 5 Note : R tot 5 reset count, cumulative total and Batch total.
Reset all (Initialisation)	Factory setting : Rst 0 Blinking 	Note : At Rst 5 reset all, all parameters reset to factory settings. Press   No reset from Rst 0 - 4 Range : Rst 0, Rst 1, Rst 2, Rst 3, Rst 4, Rst 5
NOTE : . To quit configuration mode : 1) Turn power off 2) Remove link between terminal [CONFIG] no 16, 17 for XC2200 / XC2200-C & 11, 12 for XC3200 / XC3200-C 3) Turn power on		

Config. Parameter	Display	Description
To program SET 1, SET 2 / SPBT & AR / TPR time:		
 Key : Press for 3 sec. From count mode.		
 Key : Increments the blinking digit after every key press and rolls over from 0 to 9		
 Key : Shifts the blinking to next digit (right) for every key press		
 Key : To complete programming		

Set points	Factory setting : 100 *Blinking digit  Factory setting : 90 *Blinking digit  OR Factory setting : 1 *Blinking digit  AR / TPR time *Blinking digit 	*Blinking digit  # Set value 1 Press  *Blinking digit  # Set value 2 OR Press  *Blinking digit  # Set Batch Press  *Blinking digit  # Pulse time Maximum pulse time = 99.9sec Note: 1) Time parameter is valid for AR and TPR modes. 2) SET1 should always be greater than SET2
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Count & Rate or Count & Batch display		
Count & Rate	When Rate selected in LDSP	Factory setting : Count & Rate  # Current Count value •  # Rate value with Rt (Led 3) on OR  # Current Count value •  # Batch value with Bt (Led 4) on
Count & Batch	When Batch selected in LDSP	

To Read SET 1, SET 2/ SPBT & PULSE TIME VALUES					
SET1	When SET2 is selected for Relay2 operation	Press 		#Set value 1	# Automatically skips after 1 sec
SET2		Press 		#Set value 2	
SPBT	When SPBT is selected for Relay2 operation		OR		
		Press 		#Set Batch	
TIME		Press 		#Pulse time	

TO READ VALUES OF CONFIGURED PARAMETERS :			
 Key : Read configured parameter when momentary press. # Automatically skips to value after 3 sec			
Parameters	Display Description	Parameters	Display Description
Cumulative total Batch total (momentary)	Press   # Cumulative total # Batch total with Bt (Led 4) on	Relay operating mode (momentary)	Press   # Relay operating mode
OR Cumulative total Rate (momentary)	OR Press   # Cumulative total # Rate value with Rt (Led 3) on	Reset mode (momentary)	Press   # Reset mode
Scale factor mantissa (momentary)	Press   # Scale factor mantissa	Input Frequency (momentary)	Press   # Input Frequency
Scale factor exponent (momentary)	Press   # Scale factor exponent	Rate unit (momentary)	Press   # Rate unit
Resolution (least count) (momentary)	Press   # Resolution	Overrun / non overrun mode (momentary)	Press   # Overrun
Counting mode (momentary)	Press   # Counting mode	Communication setting (momentary)	Note : Only applicable for XC2200-C & XC3200-C Press   # Baud Rate # Slave Id
Counting direction (momentary)	Press   # Counting direction	Front panel reset (momentary)	Press   # Front panel reset

MODBUS REGISTER ADDRESS LIST							
S.NO	Modbus Address	Hex Address	Parameters	Length (Bits)	Data type	R/W	Description Selection : Data
FSC,FMC,RCS							
1	00004	0 X 04	Relay Operating Mode	1	Bool	R/W	On delay : 0, Interval : 1
2	00005	0 X 05	Count direction	1	Bool	R/W	Up : 0, Down : 1
3	00006	0 X 06	Count Overflow Mode	1	Bool	R/W	Over run : 0, Non over run : 1
4	00007	0 X 07	Input Frequency	1	Bool	R/W	5000 : 0, 30Hz : 1
5	00008	0 X 08	Lower Display selection	1	Bool	R/W	Rate : 0, Batch : 1
6	00009	0 X 09	Relay 2 Operation	1	Bool	R/W	Set 2 : 0, Set Batch : 1
7	00010	0 X 0A	Front panel reset(FPR)	1	Bool	R/W	Yes : 0, No: 1
8	00011	0 X 0B	Reset all	1	Bool	R/W	Set all parameters to factory Default
9	00012	0 X 0C	Reset Batch Count	1	Bool	R/W	Reset Batch Count, Cumulative total, Count Value
10	00013	0 X 0D	Reset Count Value	1	Bool	R/W	Set Count Value to 0

S.NO	Modbus Address	Hex Address	Parameters	Length (Bytes)	Data type	R/W	Description Selection : Data
PSR, PMR, RHR							
1	40001	0 X 01	Rate unit	1	USINT	R/W	Reset per second (RPS) : 0 Rate per Minute (RPM) : 1 Rate per Hour (RPH) : 2
2	40002	0 X 02	Counting mode	1	USINT	R/W	Unidirectional : 0 Bidirectional : 1, Quad : 2
3	40003	0 X 03	Resolution	1	USINT	R/W	1 : 0 0.1 : 4 0.01 : 3 0.001 : 2 0.0001 : 1
4	40004	0 X 04	Relay Repeat Mode	1	USINT	R/W	None : 0, AR : 1, TPR : 2
5	40005	0 X 05	Scale FactorExponent	1	USINT	R/W	0 : 0 1 : 1 2 : 2 -5 : 3 -4 : 4 -3 : 5 -2 : 6 -1 : 7
6	40006	0 X 06	Slave Id		USINT	R/W	Range : 0 To 255
7	40007	0 X 07	Baud Rate	1	USINT	R/W	9600 : 2, 19K2 : 3, 38K4 : 4 57K6 : 5, 115K : 6

S.NO	Modbus Address	Hex Address	Parameters	Length (Bytes)	Data type	R/W	Description
PMR, RHR							
1	40054	0 X 36	SCL	4	UDINT	R/W	Data Format : LSB- MSB For setting Scale factor= 1.23456, send 123456 (Do not consider decimal point)
2	40056	0 X 38	SPBT	4	UDINT	R/W	Data Format : LSB- MSB For SPBT = 1, response will be: 00 01 00 00
3	40058	0 X 3A	Time(Set time value)	4	TIME	R/W	Data Format : LSB- MSB For time = 1.2 sec, send 1200 Data in decimal (set time in ms)
4	40072	0 X 48	Set1	8	LREAL	R/W	Data Format : MSB-LSB For Set1 = 100.0, response will be: 0000 4059 0000 0000H
5	40076	0 X 4C	Set2	8	LREAL	R/W	Data Format : MSB-LSB

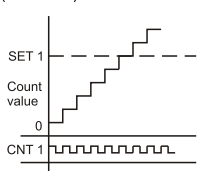
S.NO	Modbus Address	Hex Address	Parameters	Length (Bytes)	Data type	R/W	Description
RIR							
1	30014	0 X 0E	Batch Total	4	UDINT	R	Data Format : LSB- MSB
2	30016	0 X 10	Time(Run time value)	4	TIME	R	Data Format : LSB- MSB
3	30020	0 X 14	Count Total	8	LREAL	R	Data Format : MSB- LSB
4	30024	0 X 18	Cumulative Total	8	LREAL	R	Data Format : MSB-LSB
5	30028	0 X 1C	Rate	8	LREAL	R	Data Format : MSB-LSB

USER GUIDE

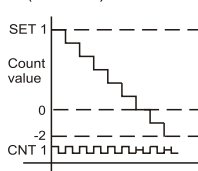
1. COUNTING DIRECTION

a. Unidirectional : The unit counts the number of pulses received at the CNT1 input and can be programmed to count upwards from zero towards the set point (**Up counting**), or to count downwards from the set point to zero (**down counting**).

Up counting
(Overrun)



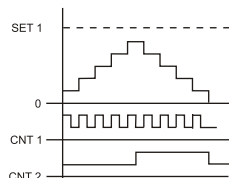
Down counting
(Overrun)



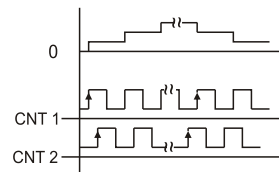
b. Bi-directional :

In the Bi-directional mode of counting, the CNT2 input determines the count direction. If the CNT2 input is at level 1 (10V to 30V) XC200-NX counts up (increments) - open terminals 3 & 4.

If the CNT2 input is at level 0 (0V to 1V) the XC200-NX counts down (decrements) - short terminals 3 & 4.



C. Quadrature : The Quadrature mode is suitable for using with sensors which generate 2 channels of output in quadrature (phase shift) e.g. Rotary encoders. The unit counts up (increments), if the CNT1 input transitions precede the CNT2 input transitions and counts down, if the CNT2 input transitions precede the CNT1 input transitions.

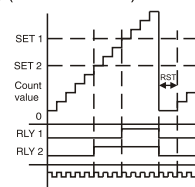


2. MODES OF OPERATION

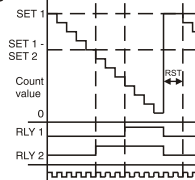
a) On delay :

The output is energised at the end of the counting cycle (i.e. count value = set value) & remains on till unit is reset. When the number of pulses received is equal to SET2, relay 2 is switched ON, and when the number of pulses received equals Set1, relay 1 is switched ON. Both the outputs remain ON till the unit is reset.

Up counting (overrun mode)



Down counting

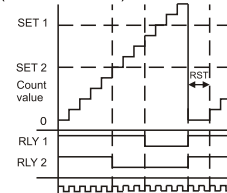


b) Interval:

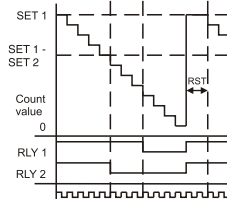
The output is energised at the start of the counting cycle and de-energised on its completion.

When the number of pulses received equal to SET2, relay2 is switched OFF, and when the number of pulses received equal to Set1, relay 1 is switched OFF. These relays remain OFF till the unit is reset.

Up counting (overrun mode)



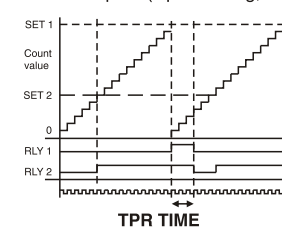
Down counting



c) Time Pulse Repeat (TPR) :

The output changes state (according to the On delay or Interval mode selected) for a programmed pulse time (t), at the end of the count cycle. After the set pulse time, the count resets automatically & count cycle repeats. Counting continues uninterrupted during the pulse time.

Time Pulse Repeat (Up Counting, On Delay)

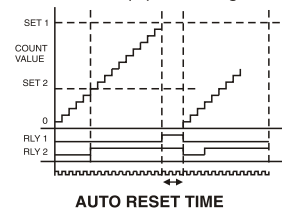


d) Auto-reset mode :

The output energises (according to the On delay or Interval delay mode selected) at the end of the count cycle for the programmed Auto reset time, after which the counter automatically resets and begins the next count cycle.

The unit remains reset for the Auto reset time and ignores count pulses received during this duration. It resumes operation only after this period has elapsed.

Auto Reset Mode (Up Counting, On Delay)



Overrun / Non Overrun feature:

Overrun: In overrun mode, the counter continues counting, after the SET1 value has been crossed.

Non overrun: In non overrun mode, the counter stops counting, after the SET1 value is reached and restarts counting only after reset.

SCALE FACTOR

The user programmable scale factor facilitates the direct reading in desired engineering unit.

The counter multiplies the number of pulses received at the count input with the scale factor, and displays the result.

Count display = No. of pulses received x scale factor

Rate display = Number of pulses received per minute x scale factor or No. of pulses received per hour x scale factor.

The scale factor consists of two parts, mantissa and exponent.

The mantissa can be set from 0.00001 to 9.99999 and the exponent can be set from -5 to +2. The scale factor is arrived at as:

Scale factor = Mantissa X 10^{Exponent}

(Specifications are subject to change, since development is a continuous process.)

Selec Controls Pvt. Ltd., India

Factory Address :

EL-27/1, Electronic Zone, TTC Industrial Area, MIDC, Mahape, Navi Mumbai - 400 710, INDIA.

Tel. No. : +91-22-41 418 419/430 | Fax No. : +91-22-28471733

Toll free : 1800 227 353 (BSNL/MTNL Subscribers only)

Website : www.selec.com | Email : sales@selec.com

In Overrun mode, the unit continues to count above the set point.